Low Acrylonitrile NBR

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Abstract

This report synthesizes patent data pertaining to the synthesis of Low Acrylonitrile Nitrile Butadiene Rubber (NBR), focusing on the polymerization processes and key reaction parameters. The analyzed patents consistently employ emulsion polymerization as the primary synthesis route for NBR, indicating its established industrial relevance for producing these elastomers. The core monomeric components across all patents include conjugated dienes, specifically butadiene, and alpha,beta-unsaturated nitriles, with acrylonitrile being the defining component for NBR. Some patents also incorporate carboxylic acids (e.g., methacrylic acid) or other copolymerizable monomers (e.g., methyl isopropenyl ketone) to modify polymer properties, suggesting a strategy to tailor the final material's characteristics.

A notable trend is the emphasis on controlling the acrylonitrile content, with one patent (US9926439) explicitly defining it between 17 and 45% by weight, alongside methacrylic acid content of less than 15%. This highlights the importance of monomer ratios in achieving specific material attributes, particularly those mimicking natural rubber characteristics. While specific emulsifier selections are consistently not disclosed, the use of redox initiator systems is mentioned in one patent (EP2316860), comprising a peroxo compound, a reducing agent, and iron(II)chloride, indicating a preference for controlled radical polymerization. Chain transfer agents, such as C12-C16-alkylthiols, are also identified as crucial for molecular weight regulation.

Reaction conditions vary, with temperatures ranging from 30 °C to 85 °C and reaction times from 15 minutes to 72 hours, suggesting diverse approaches to achieve desired conversions and polymer structures. However, detailed information on pressure, pH, and conversion rates is largely absent. Post-polymerization steps like coagulation and specific water amounts are also generally not disclosed, limiting a comprehensive understanding of the full process chain. Overall, the patent landscape for Low Acrylonitrile NBR synthesis underscores the reliance on emulsion polymerization, the strategic selection of monomers and initiators, and the critical role of reaction conditions in tailoring polymer properties, despite a general lack of granular detail in many disclosed parameters.

Methodology

EP2507313 - Elastomeric rubber and rubber products without the use of vulcanizing …

- **Polymerization Process:** emulsion polymerization
- **Monomers:** butadiene, acrylonitrile, carboxylic acid
- **Monomer Ratio:** Not disclosed
- **Initiator:** Not disclosed
- **Emulsifier:** Not disclosed
- **Catalyst:** Not disclosed
- **Chain Transfer Agent:** Not disclosed
- **Coagulation:** Not disclosed
- **Water Amount:** Not disclosed
- **Reaction Temperature:** 85 °C
- **Reaction Time:** 72 hours
- **Reaction Pressure:** Not disclosed
- **Reaction pH:** Not disclosed
- **Conversion:** Not disclosed
- **Solid Content:** 18 %
- **Mooney Viscosity:** Not disclosed
- **Volatile Matter:** Not disclosed
- **Ash:** Not disclosed
- **Other Properties:** Not disclosed
- **Reaction Procedure:** Not disclosed
- **Experimental Notes:** Not disclosed

US9926439 - Nitrile rubber article having natural rubber characteristics

- **Polymerization Process:** emulsion polymerization
- **Monomers:** acrylonitrile, methacrylic acid, butadiene
- **Monomer Ratio:** acrylonitrile content between 17 and 45% by weight, methacrylic acid content of less than 15% by weight, remaining balance butadiene
- **Initiator:** Not disclosed
- **Emulsifier:** Not disclosed
- **Catalyst:** Not disclosed
- **Chain Transfer Agent:** Not disclosed
- **Coagulation:** Not disclosed
- **Water Amount:** Not disclosed
- **Reaction Temperature:** Not disclosed
- **Reaction Time:** 15 minutes
- **Reaction Pressure:** Not disclosed
- **Reaction pH:** Not disclosed
- **Conversion:** Not disclosed
- **Solid Content:** 15-25% total solids content

- **Mooney Viscosity:** Not disclosed
- **Volatile Matter:** Not disclosed
- **Ash:** Not disclosed
- **Other Properties:** Not disclosed
- **Reaction Procedure:** Not disclosed
- **Experimental Notes:** Not disclosed

EP2316860 - Nitrile rubbers

- **Polymerization Process:** emulsion polymerization
- **Monomers:** at least one alpha,beta-unsaturated nitrile, at least one conjugated diene and optionally at least one further copolymerizable monomer (e.g., 1,3-butadiene, acrylonitrile, methyl isopropenyl ketone)
- **Monomer Ratio:** Not disclosed
- **Initiator:** redox system comprising (a) at least one peroxo compound as an oxidizing agent, (b) at least a reducing agent and (c) iron(II)chloride
- **Emulsifier:** Not disclosed
- **Catalyst:** Not disclosed
- **Chain Transfer Agent:** C12-C16-alkylthiol
- **Coagulation:** Not disclosed
- **Water Amount:** Not disclosed
- **Reaction Temperature:** 30 °C
- **Reaction Time:** 4 hours
- **Reaction Pressure:** Not disclosed
- **Reaction pH:** Not disclosed
- **Conversion:** Not disclosed
- **Solid Content:** Not disclosed
- **Mooney Viscosity:** Not disclosed
- **Volatile Matter:** Not disclosed
- **Ash:** Not disclosed
- **Other Properties:** Not disclosed
- **Reaction Procedure:** Not disclosed
- **Experimental Notes:** Not disclosed

Cross-Patent Comparison & Synthesis Trends

The analysis of the provided patent data reveals several common technical trends in the synthesis of Low Acrylonitrile NBR, alongside variations in specific parameters.

Emulsifier Selection and Loading

Across all three patents (EP2507313, US9926439, EP2316860), the specific emulsifier used for the emulsion polymerization process is consistently "Not disclosed." This lack of disclosure suggests that while emulsion polymerization is the universally adopted method, the specific emulsifier chemistry might be considered proprietary or standard practice within the field, thus not warranting explicit mention in the provided summaries. Without this information, it is difficult to infer trends in emulsifier type, concentration, or their impact on particle size and stability.

Initiator and Chain Transfer Agent Strategies

Initiator strategies show some variation. While EP2507313 and US9926439 do not disclose their initiators, EP2316860 explicitly details the use of a "redox system comprising (a) at least one peroxo compound as an oxidizing agent, (b) at least a reducing agent and (c) iron(II)chloride." This indicates a preference for controlled radical polymerization, which is common in emulsion systems to manage reaction rates and polymer properties. The use of iron(II)chloride as a co-catalyst in the redox system is a well-known approach to enhance initiation efficiency.

Regarding chain transfer agents (CTAs), EP2316860 specifies the use of a "C12-C16-alkylthiol." Alkylthiols are widely employed in NBR synthesis to control molecular weight and molecular weight distribution, thereby influencing the processability and final mechanical properties of the rubber. The other two patents do not disclose their CTA, but it is highly probable that some form of CTA is used to manage polymer chain length.

Monomer Ratio and Reaction Control

All patents utilize butadiene and acrylonitrile as primary monomers, with the addition of a third monomer for property modification. EP2507313 mentions "carboxylic acid," and US9926439 specifies "methacrylic acid," while EP2316860 broadly refers to "at least one further copolymerizable monomer" such as "methyl isopropenyl ketone." This trend indicates a common strategy to introduce functional groups or modify the glass transition temperature and polarity of the NBR.

US9926439 provides the most specific monomer ratio, stating "acrylonitrile content between 17 and 45% by weight" and "methacrylic acid content of less than 15% by weight, remaining balance butadiene." This range for acrylonitrile clearly defines the "Low Acrylonitrile NBR" category and highlights the importance of precise monomer feed ratios in achieving desired material characteristics, particularly those mimicking natural rubber. The other patents do not disclose specific monomer ratios, making it challenging to draw further comparisons on this aspect.

Reaction conditions vary significantly. Temperatures range from 30 °C (EP2316860) to 85 °C (EP2507313), and reaction times span from 15 minutes (US9926439) to 72 hours (EP2507313). This wide range suggests different polymerization kinetics and strategies, potentially aimed at varying conversion rates, molecular weights, or microstructures. However, detailed information on pressure, pH, and conversion is consistently "Not disclosed," limiting insights into the precise control mechanisms employed during polymerization. Solid content is reported between 15-25% for US9926439 and 18% for EP2507313, indicating typical latex concentrations for emulsion polymerization.

Coagulation and Post-Treatment Conditions

Information regarding coagulation and post-treatment conditions is uniformly "Not disclosed" across all analyzed patents. This suggests that these steps, while critical for isolating the polymer from the latex and preparing it for subsequent processing, are either considered standard industrial practices or are not deemed central to the novelty claimed in these specific patents. Without this data, it is not possible to identify any specific trends or innovations in polymer recovery and purification for Low Acrylonitrile NBR.

References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
EP2507313	Elastomeric rubber and rubber products without the use of vulcanizing …	Not disclosed	EP	2018-02-07	EP2507313B1
US9926439	Nitrile rubber article having natural rubber characteristics	Not disclosed	US	2018-03-27	US9926439B2
EP2316860	Nitrile rubbers	Not disclosed	EP	2012-05-09	EP2316860B1